Matrices exercises

All questions are original. Unless stated otherwise, entries are real, dimensions are positive integers, and arithmetic is exact. Rank counts independent columns, rather than comparing floating-point entries with a threshold.

D-E01 Dimensions and entrywise multiplication

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}.$$

Compute every entry of AB and BA, and decide whether the products can be equal. For $x = (1, -1, 2)^T$, compute Ax using both row dot products and a linear combination of columns.

D-E02 Associativity and noncommutativity

Prove $(AB)C = A(BC)$ using finite sums, and state the compatible dimensions. Give two real symmetric 2×2 matrices whose product is not symmetric. For real symmetric matrices A and B of the same size, prove that AB is symmetric if and only if $AB = BA$.

D-E03 Permutation direction

Let

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}.$$

Find PM, MP, and P^T, and verify $P^T P = I_3$. Explain why left and right multiplication reorder the entries in different directions.

D-E04 Rank and the whole null space

For

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix},$$

show every Gauss–Jordan step. Form B from the original pivot columns and find the coordinate matrix C such that $M = BC$. Give a basis for the null space, the rank, and the nullity. Explain why the pivot columns of the reduced row echelon form (RREF) cannot directly replace the original columns in B.

Matrices exercises (continued)

D-E05 Product rank and invertibility

Prove $\operatorname{rank}(AB) \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$. Give rank-one matrices A and B with $AB = 0$. Prove $\operatorname{rank}(PA) = \operatorname{rank}(A)$ when P is invertible, and state the dimension and field assumptions.

D-E06 Complete inverse trace

Find the inverse of $H = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ by augmenting it with the identity. Show every intermediate augmented matrix and check the result by multiplication on both sides. Prove uniqueness of the inverse and give a rectangular matrix with a one-sided inverse that is not an inverse on the other side.

D-E07 Cofactors and determinant rules

For $N = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & -1 & 1 \end{bmatrix}$, list the first-row minor matrices, cofactors, and determinant. Recompute the determinant after adding the first row to the second, and then after swapping the first two rows. Prove the diagonal-product rule for triangular matrices. State the assumptions under which $\det A = 0$ is equivalent to singularity.

D-E08 Outer-product boundaries

Let $u = (1, 2)^T$ and $v = (3, -1)^T$. Compute $u^T v$, uv^T, and the rank of uv^T, and then repeat with $u = 0$. Prove that an outer product has rank at most one, with equality when both vectors are nonzero. Over $\mathbb{C}$, explain how the outer product uv^* and the inner product change.

D-E09 Gram and field counterexamples

Compute the Gram matrices for the lecture matrices

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad W = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}.$$

Use a general vector $(s, t)^T$ to prove positive definiteness, and use a nonzero null vector to demonstrate failure of strict positivity. Compare $A^T A$ and $A^* A$ for the complex matrix $A = [i]$. Identify the two missing assumptions in the claim that a nonzero A implies positive definiteness of $A^T A$.

Matrices exercises (continued)

D-E10 GF(2) maps and representation

Over $\mathbb{F}_2$, let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. The bit vector $(x_0, x_1)^T$ represents the integer $x = x_0 + 2x_1$. List the integer outputs for $x = 0, 1, 2, 3$, and find the rank and inverse of A. Explain why real arithmetic cannot be used unchanged. Why does a nonzero pivot 2 fail to permit division modulo 6?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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